

Rationalization of Thermal Simulators for Operations - Real Time Thermal Flight Correlation

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This paper focuses on the rationalization of thermal simulators for operational purposes, leveraging existing thermal models and tools. By reusing thermal mathematical models throughout the satellite's lifecycle, spanning design, thermal analysis, testing, and operations, this approach aims to enhance efficiency and reduce costs. Sharing a common thermal model between analysis and operations enables spacecraft manufacturers, such as Thales Alenia Space, to streamline simulator implementation while improving accuracy. The study uses Sentinel-3 as a case study due to its flight status and well-characterized platform. The activity, supported by ESA and CNES, involved implementing a real-time spacecraft thermal simulator demonstrator based on Sentinel-3B's thermally validated model from thermal vacuum testing. This simulator addresses in-flight operational challenges such as anomaly detection and recovery testing by simulating new active thermal control strategies. Key technical challenges include telemetry recovery, initializing thermal model temperatures from telemetry, and achieving real-time performance. The simulator integrates a dynamic orbit propagator, PI-based thermal regulation, and a platform dissipator, while solving radiative and thermal equations on-the-fly even for unpredicted scenarios. The present paper details the current results of the real-time thermal flight correlation study, including replay simulations (8x faster than real time) that demonstrate performance for a hot-case scenario validated against telemetry. Additionally, the benefits of a reliable thermal simulator are discussed, including the ability to validate thermal behavior, refine margins, and correlate models during testing and flight. Potential future work, notably estimating equipment dissipation from telemetry and expanding use-case demonstrations, are detailed.

Acronyms and Nomenclature

AGYR	=	Automatic Generated Yearly Report
AIT	=	Assembly, Integration and Testing
CNES	=	French National Centre for Space Studies
DSS	=	Dynamic Spacecraft Simulator

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<i>EOR</i>	= <i>Electric Orbit Raising</i>
<i>ESA</i>	= <i>European Space Agency</i>
<i>GMM</i>	= <i>Geometric Mathematical Model</i>
<i>GPP</i>	= <i>Graphic Post Processing</i>
<i>GUI</i>	= <i>Graphic User Interface</i>
<i>INRIA</i>	= <i>French National Institute for Research in Digital Science and Technology</i>
<i>LEOP</i>	= <i>Launch and Early Orbit Phase</i>
<i>MPL</i>	= <i>Mechanically Pumped Loop</i>
<i>MUST</i>	= <i>Mission Utility and Support Tools</i>
<i>MWR</i>	= <i>Micro-Wave Radiometer</i>
<i>OBS</i>	= <i>On Board Software</i>
<i>OLCI</i>	= <i>Ocean and Land Color Instrument</i>
<i>OME</i>	= <i>Opto-Mechanical Enclosure</i>
<i>P/F</i>	= <i>Platform</i>
<i>PI</i>	= <i>Proportional Integral</i>
<i>P/L</i>	= <i>Payload</i>
<i>POC</i>	= <i>Proof Of Concept</i>
<i>PSR</i>	= <i>Post-Simulation Review</i>
<i>S3B</i>	= <i>Sentinel 3B satellite</i>
<i>SA</i>	= <i>Solar Arrays</i>
<i>SLSTR</i>	= <i>Sea Land Surface Temperature Radiometer</i>
<i>SRAL</i>	= <i>Synthetic Aperture Radar Altimeter</i>
<i>TAS</i>	= <i>Thales Alenia Space</i>
<i>TM</i>	= <i>Telemetry</i>
<i>TMM</i>	= <i>Thermal Mathematical Model</i>
<i>TRL</i>	= <i>Technology Readiness Level</i>
<i>TRP</i>	= <i>Temperature Reference Point</i>

I. Introduction

The advancement of space technologies relies heavily on the development of efficient and reliable thermal management systems, which are critical for ensuring the optimal performance and longevity of satellites. Funded and supported by the French Space Agency (CNES) and technically led by the European Space Agency (ESA), this initiative aimed to address the challenges associated with thermal simulation in satellite operations, notably regarding real-time simulation. The project was executed by a tendering team, with Dorea as the prime contractor and Thales Alenia Space as a key subcontractor, leveraging their expertise, models, and data, particularly for the Sentinel-3B satellite of the Copernicus constellation. The primary objective of this project was to achieve a proof of concept (Technology Readiness Level 3, TRL3) for the rationalization of real-time thermal simulators. This was accomplished through the development of a demonstrator based on a correlated Thermal Mathematical Model (TMM) following Post-Simulation Review (PSR). The specific use case focussed on satellite operations, emphasizing the comparison between simulation outputs and telemetry data. By streamlining thermal simulation processes, this project aimed to enhance the accuracy and efficiency of thermal predictions, ultimately contributing to improved mission planning and operational reliability for future space missions. The methodology adopted in this study was based on the state-of-the-art in thermal simulation, which primarily revolves around two approaches: *Surrogate Models*^{1,2,3} (with or without artificial intelligence) and *TMM*. TAS has pioneered the use of reduced-order models, such as those derived from the EQUIVAL method^{4,5}, to achieve smaller yet accurate thermal models. These numerical methods have demonstrated their capability to meet real-time constraints and beyond, as evidenced by their successful implementation in projects like O3B and Alphasat at TAS Cannes for the DSS since 2012. Furthermore, TAS employs TMMs correlated during PSR to ensure high fidelity in thermal predictions.

The *Surrogate Model* approach, particularly when enhanced with AI, is well-suited for addressing complex physical phenomena that lack straightforward mathematical representations, involve a limited number of observables, and exhibit reduced variability. However, the tool developed in this study was designed to be modular, ensuring adaptability to diverse operational scenarios. In addition, the tool is modular since it can interface with multiple simulation software, such as ESATAN-TMS, eTherm, Systema-Thermica and others. Importantly, these methodological constraints were tailored to the specific needs of satellite operations rather than the broader satellite

design phase, aiming to lean towards a thermal digital twin. Regarding the latter, recent works have been proposed for thermal applications, notably for energy efficiency in buildings⁶ or for the modelisation of thermal power plants⁷.

II. Digital Twin Choice

A. Background

This first activity focussed on the rationalization of thermal simulators for operations. These thermal simulators are based on correlated PSR-level thermal models and existing thermal analysis tools. This activity was complementary to an ESA R&D activity "Towards a thermal digital twin" by proposing new methodologies and tools to support thermal design, development, testing and operations. This activity was a feasibility study on the implementation of a thermal digital twin connected to TM flight with the objective of the use-case related to operations and after-sales service and with the scientific satellite S3B as the application program.

This activity drew on the expertise and know-how from past and ongoing studies, being the development of a real-time thermal simulator for operations dedicated to Telecom satellites since 2012, the development of a thermal payload simulator for the customer dedicated to Telecom satellites since 2016, the STEP-TAS/NRF for the exchange of thermal model data, the co-simulation between eTherm and commercial fluidic tools (AMESIM), as well as a thesis on the reduction of the radiative geometric model by Artificial Intelligence in collaboration with INRIA laboratory, with a tool allowing the reduction of a geometric model respecting physical constraints (physics-informed)⁸.

The industrial team, led by DOREA and Thales Alenia Space France, has demonstrated all the expertise to solve the technical challenges of this project and all the experience on the Sentinel-3B satellite which was the application program for this activity, see Figure 1 for an overview of S3B. DOREA and TAS have extensive experience and expertise in the development of real-time thermal simulators. In addition, DOREA funded a doctoral thesis on the reduction of radiative geometric models by Artificial Intelligence in collaboration with INRIA, a thesis successfully defended in June 2022, and proposed to use the tool implemented in the exercise of reducing the radiative geometry of the thermal model.

Finally, this activity corresponded to the establishment of a digital twin since the objectives coincided with those described by the American analysis firm Gartner, namely a real-time update by the observed data.

B. TMM or Surrogate Model Approach

The real-time constraint requires the use of fast tools based on optimized algorithms. Experience in providing thermal simulators for operations has clearly identified that the effort must be placed on the radiative calculation module. Indeed, solving the heat equation is an operation that is in line with real-time calculations and even beyond, whether with the ESATAN-TMS or eTherm solver. On the other hand, with regard to radiative calculation, we have shown initially that a module based on a deterministic calculation method (eTherm: direct estimation of the quadruple integral formulating a view factor) is faster (factor 5 to 10) than a module based on a probabilistic calculation method (ESATAN-TMS: Monte Carlo method, evaluation of a view factor by ray tracing). But this first choice is not enough, it was necessary to work, on TAS funding, on algorithmic optimization, in particular on the parallelization of radiative calculations but also on the storage of radiative paths so that the consideration of the aging of the satellite can be deduced by a "simple" matrix multiplication between the thermo-optical property and the path. Furthermore the deterministic calculation method regarding view factor computations is being adapted for GPU execution, and the first outcomes are encouraging, early results showing a speedup by a factor 3 on a small GPU compared to the fully parallelized CPU method. This approach

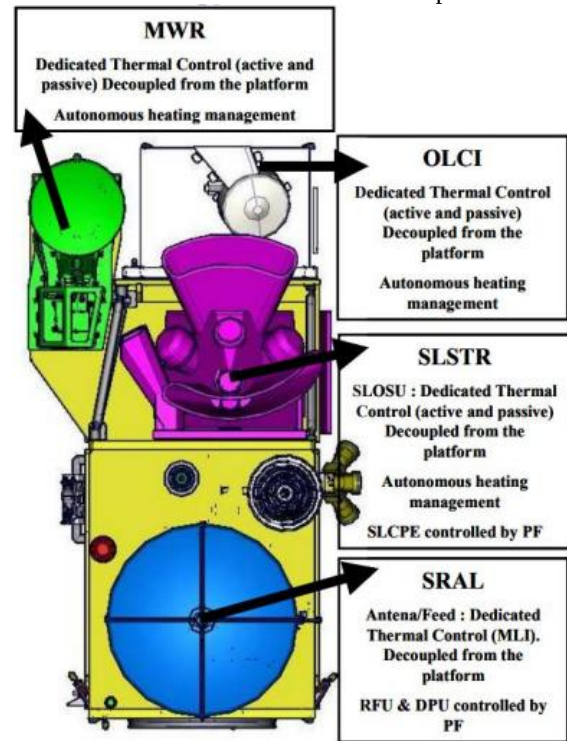


Figure 1. Sentinel-3 Satellite Overview. Dedicated Thermal Control entities are depicted.

justifies the use of a TMM instead of a surrogate model type approach whose current maturity remains to be demonstrated. However since the Digital Twin is designed to be modular, a mature surrogate model solution could be used instead of the TMM approach.

III. Use Cases and Technological Choices

A. Use Cases

The use cases for a thermal digital twin span multiple phases of a satellite's lifecycle. During the *design phase*, a thermal digital twin could be essential for thermal design conception, including orbit and attitude configuration analysis, thermo-energetic assessments, and the evaluation of 'what-if' scenarios. In the *post-design phase*, main advantages of such a digital twin lie in the initialization of models from on-flight telemetry, correlation models with actual flight data (e.g., dissipation parameters), detection of anomalies, and to provide additional observables from the TMM.

In the AIT phase, a thermal digital twin seeks to support the optimization of test sequences, reduce test bench time, optimize thermocouple distribution, and facilitate data exploitation between thermal balances, with a growing emphasis on real-time correlation. Finally, during the *operations phase*, such thermal digital twin is critical for monitoring aging effects, analyzing cryocooler behavior, planning missions, configuring payloads, detecting energetic anomalies, optimizing battery usage, estimating fuel consumption, and conducting deorbit and re-entry analyses.

Another considered use case of such a digital twin consists in the usage of the real-time thermal cartography of the satellite as well as infrared cameras for the docking phase of an entity onto another one. This is part of the RAPTOR project between IRT Saint-Exupéry and TAS.

B. Technological Choices

The software architecture is designed to be modular, divided into a *frontend* and a *backend*. This architecture is designed to ensure user-friendly interaction, secure data exchange, and efficient management of simulation processes.

The frontend, including the GUI, is built using *Angular JS* (with *Node JS*), providing an intuitive and responsive interface for users. The GUI is virtualized to ensure cross-platform compatibility and seamless user interaction. Secure communication between the frontend and backend is achieved through *JSON Web Tokens (JWT)*, encrypted using *SHA-256* for enhanced security. The connection is established via *TCP-IP* endpoints, with *OAT++* and *JWT* serving as connectors to facilitate secure and efficient data exchange. This architecture ensures that users can interact with the system securely while maintaining real-time responsiveness.

The backend MOD is the core of the system, responsible for managing user roles, simulation configurations, and data processing. The system supports role-based access control, distinguishing between *operators* and *experts*. Operators can configure and monitor simulations, while experts have additional privileges to modify model parameters and analyze complex scenarios. The backend integrates with online and replay telemetry systems, overlaying the *AGYR software* for telemetry recovery and workbench management. This allows users to initialize simulations from actual flight data, ensuring high fidelity and accuracy. The system enables the creation of simulation instances based on *S3B templates*. Users can edit TMM characteristics such as node definitions and material properties, and thermal regulation characteristics including heater control logic and setpoints. A comprehensive monitoring system is implemented to track simulation progress and detect anomalies. Alarms are triggered based on predefined thresholds, ensuring timely intervention and analysis. To account for the long-term effects of space environments, the system models thermo-optical properties as a function of time. This feature enables users to simulate and analyze the impact of aging on satellite performance over its operational lifespan. Simulations can be initialized using actual telemetry data, ensuring that the models are correlated with real-world conditions. This capability is critical for anomaly detection, diagnosis, and mission planning. The system is designed to be modular, allowing for easy integration of new features and adaptation to different satellite platforms or mission requirements. This flexibility ensures that the tool can be extended beyond the specific use case of Sentinel-3B.

The *SIM* thermal simulator is a tool designed to overcome the limitations of the *OBS* system. *SIM* has the ability to modify input files *on the fly* during simulation, for instance the Keplerian orbit parameters, maneuver profiles, thermal regulation thresholds, and equipment dissipations. This flexibility allows operators to adapt simulations in real time, enabling rapid scenario testing and anomaly resolution. The simulator incorporates an orbit propagator to accurately predict the satellite's position and orientation over time. This is critical for calculating external thermal loads, such as solar flux and Earth albedo. A kinematics propagator is included to model the movement of satellite components, such as solar arrays. This ensures that the thermal impact of dynamic elements is accurately captured in

the simulation. The thermal regulation system in *SIM* uses a *PI* control algorithm⁹, scheduled at a duty cycle of 32 seconds. The dissipator module simplifies the calculation of equipment dissipations by modeling them as a function of telemetry parameters, such as orbit time, temperature, or equipment status.

The *TCS* simulator, already in use for telecom satellite operations, complements the *SIM* simulator. *TCS* can connect to *SIM* or other legacy systems (e.g., *OBS*) via *TCP-IP*, enabling seamless integration with existing infrastructure. This ensures compatibility and facilitates data exchange between different simulation environments. The core of *TCS* lies in its ability to solve the heat equation efficiently, *TCS* also calculates absorbed solar, albedo, and Earth radiative powers in parallel, significantly reducing computation time. The simulator provides instantaneous calculations of average and extreme temperatures, as well as power balances.

IV. Software Overview

A. Telemetry Acquisition and Output Interfaces with AGYR

To ensure seamless integration with existing operational systems, the thermal simulator incorporates dedicated interfaces for both telemetry acquisition and output, as well as simulated data exchange. TM acquisition is managed through two primary sources: TM packet files retrieved from the MUST system and individual TM parameter acquisition via the Web MUST interface. On the output side, TM data can be accessed through a dedicated TM parameters request API, enabling efficient retrieval of relevant telemetry information. Additionally, simulated data is handled via structured interfaces, including a file storage API for data archiving and a file request API for controlled access to stored simulation results. These interfaces ensure efficient data flow between the simulation environment and operational tools, enhancing the accessibility and usability of the thermal simulator for real-time and post-processing applications. An overview of the software architecture can be seen on Figure 2, regarding AGYR more details can be found Figure 3.

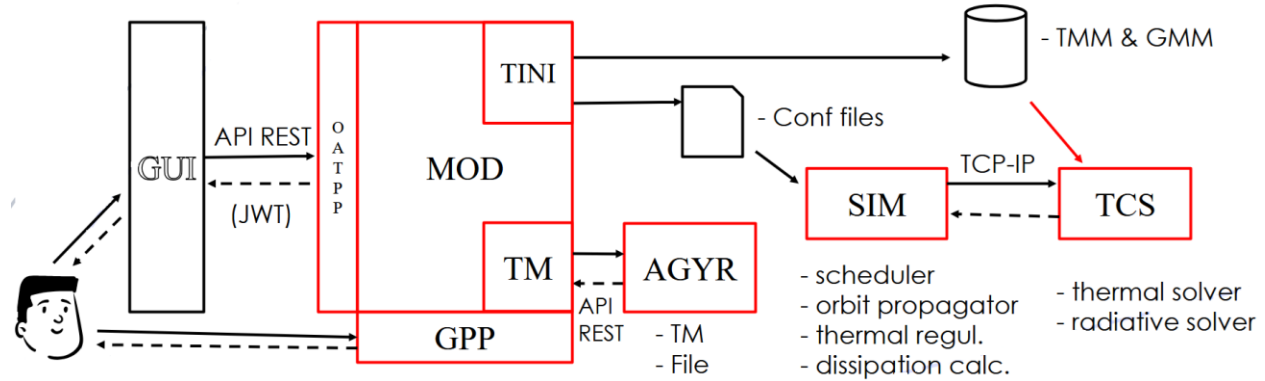


Figure 2. Software Architecture. Modules and communication protocols are depicted.

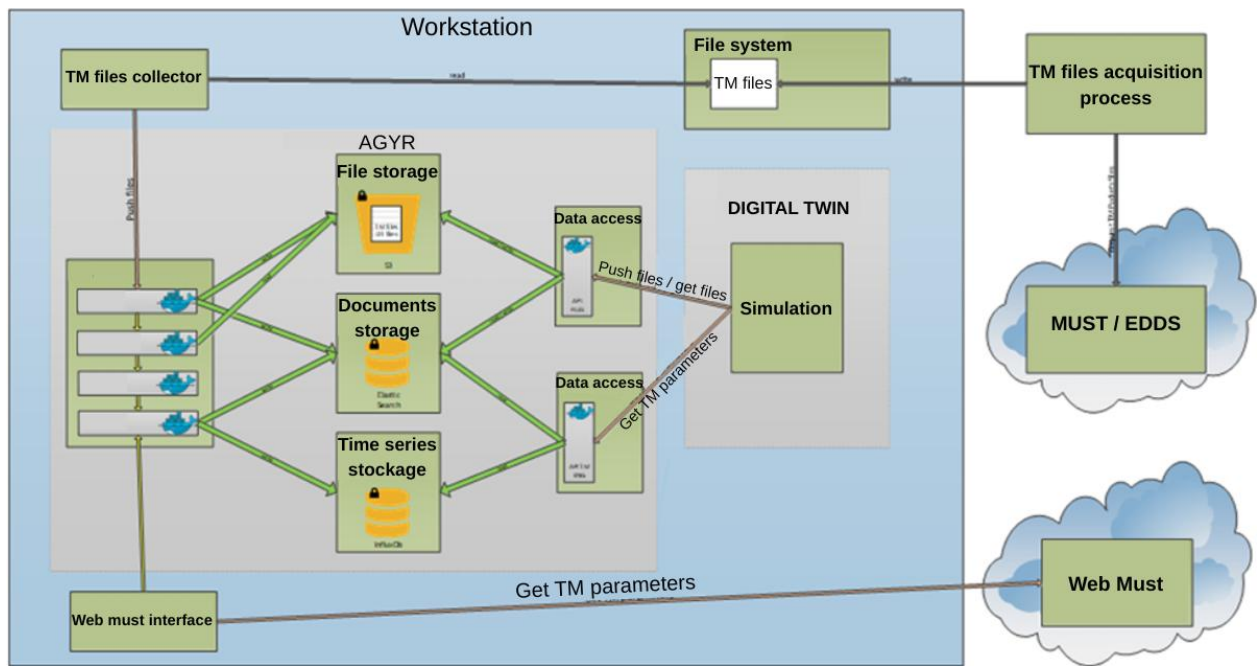


Figure 3. AGYR Overview. TM files acquisition workflow is depicted.

B. Thermal Digital Twin Software Overview

The frontend of the rationalized thermal simulator is designed to provide a secure and user-friendly interface (see Figure 4), accessible via a URL endpoint with JWT authentication (SHA-256 coupled to an internal encodement). It is fully compatible with modern web browsers, including Firefox, Chrome, and Microsoft Edge. The platform supports



Figure 4. Thermal Digital Twin Main View. Thermal nodes visualization is depicted.

different user roles, expert and operator, offering contextual features and views tailored to each profile. User

management is handled through an administrative expert account, ensuring controlled access and role-based functionality.

The system enables satellite selection via the AGYR database, allowing users to define relevant URL parameters for data retrieval. Workbenches provide an organized environment for managing different simulation setups, including TMM configuration and the definition of thermal regulation parameters. Users can specify multiple simulation runs with predefined threshold settings, ensuring flexibility in testing and analysis. The platform supports real-time monitoring and enables simulations in three modes: real-time, replay, or predictive runs. Additionally, initialization from telemetry data is possible, where temperature thresholds for TRP guide the starting conditions. Finally, the system allows for maneuver definition, further enhancing its applicability for operational planning and analysis, see Figure 5 for a global view of the main UI functionalities.

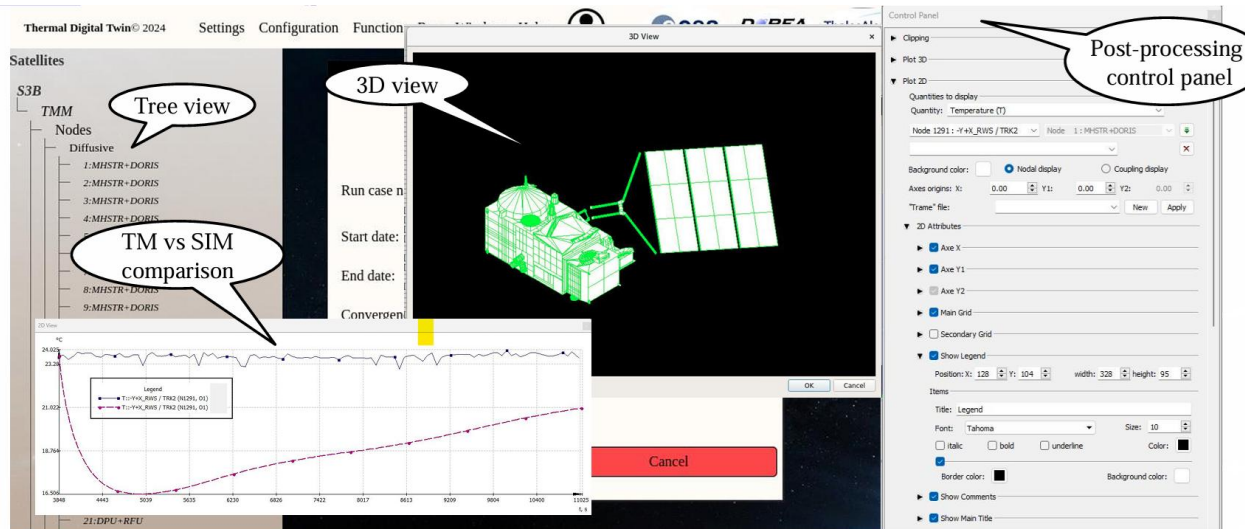


Figure 5. Global Thermal Digital Twin UI Functionalities. 2D and 3D views are depicted.

V. Performances and Results

A. Performances

The simulations were run on an off-the-shelf PC with respectable computing power. The physical time constant of the S3B TMM is 0.1 seconds, a constraint imposed by the presence of numerous small thermal nodes. The model consists of 4,051 thermal nodes, and the satellite's orbit duration, as measured from telemetry data, is 6,048 seconds. Each orbit simulation takes 753 seconds to complete, making it approximately eight times faster than real-time. The number of duty cycles is 189 and the GMM contains 70,000 external geometric faces.

Regarding the SIM component, orbit propagation per cycle requires 0.26 seconds, while dissipator calculations take 0.35 seconds and are executed in parallel with the SIM process.

For the TCS component, radiative calculations for 32,000 faceted elements, after geometric reduction, require 2.74 seconds. The resolution of the heat equation takes 3.98 seconds, with potential improvements possible by simplifying the TMM. The heat equation resolution duration per step is 0.1 seconds.

The real-time constraint lock has been lifted notably thanks to radiative computations acceleration through parallelization and geometric model reduction.

B. Current Results from TM Correlation

One of the key features of this project was to demonstrate the feasibility of comparing data collected from telemetry with data issued from simulation in real-time, notably through dynamic 2D plots. The following section describes the correlation with the hot reference case. Regarding the latter, the date is set to 13th of February 2024, the simulation duration is set from 00h00 until 04h00 am. As for the initialization, the nodes associated with TM are initialized with the actual flight temperatures, the other nodes temperatures are initialized by simulating a quarter of an orbit. The instruments nodes and interfaces nodes associated with TM are declared as classic nodes in this case. To ensure representativity of the results, the temperatures of these nodes must be imposed with the actual TM temperatures.

Figure 6 and Figure 7 compare the temperature results from the simulation in blue for a given node with the associated TM temperatures in red.

These early results are encouraging, as the telemetry and simulation exhibit the same overall shape, with only a slight offset along the time and the temperatures (about two Celsius degrees) between them. The offsets between TM and simulated data could be explained from the technological locks, for instance the temperature offset could result

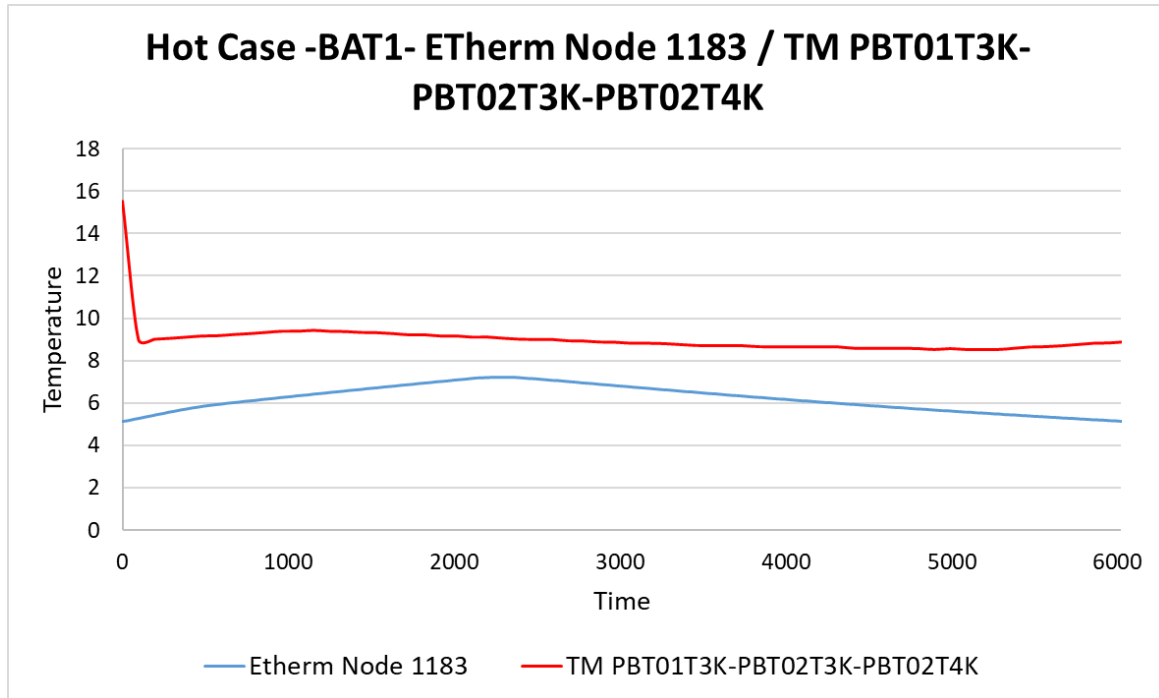


Figure 6. Correlation between a node of Digital Twin TMM and the corresponding TM. Here the node/TM is associated to the battery of the satellite.

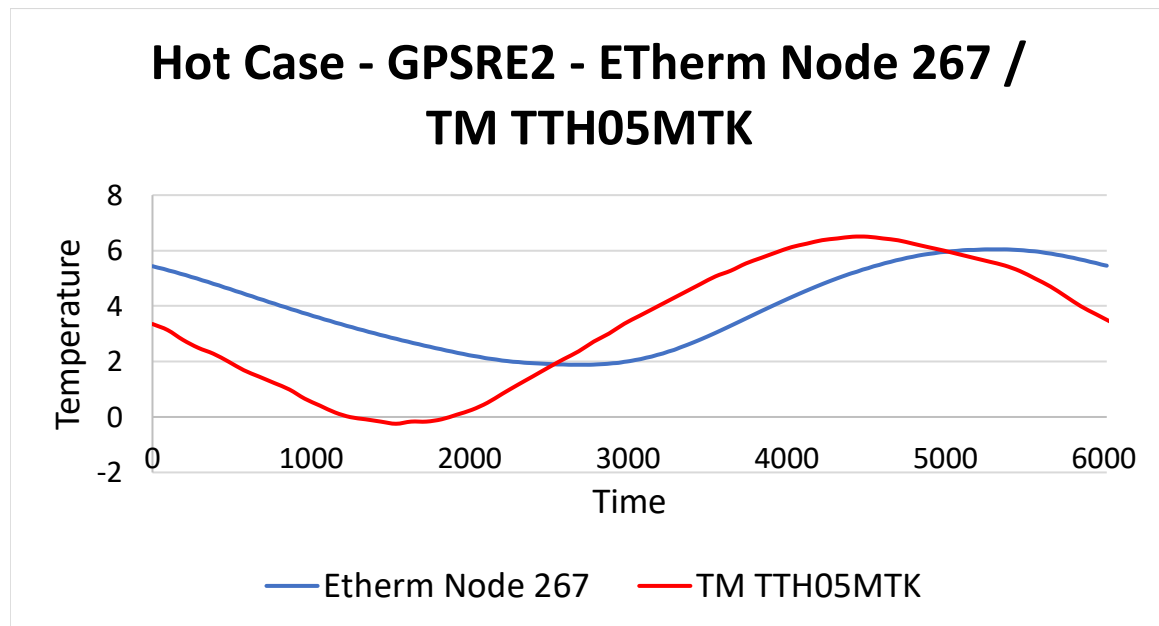


Figure 7. Correlation between a node of Digital Twin and the corresponding TM. Here the node/TM is associated to the GPSRE2.

from the inertia of the satellite, meaning that several simulated orbits are required in order to fit to the TM data. These technological locks shall be studied in the near future.

C. Application to Reliability Computations

As a consequence of the acceleration of radiative computations, the *SIM* and *TCS* thermal simulators have been rigorously validated through their application in a high-fidelity reliability analysis of a TAS telecommunication satellite. The objective was to compute the average temperatures of thermal nodes associated to the TRP, these quantities being involved for the determination of the insurance cost of the satellite. This case study demonstrates the capabilities of the simulators in handling complex, long-duration thermal simulations while accounting for critical factors such as aging, eclipse effects, and thruster operations.

The primary objective of this case study was to assess the satellite's thermal reliability over its entire operational lifespan, which spans **17 years (5,844 days)**. The simulation incorporated the effects induced by the aging of materials, the eclipses and the thrusters. Thermo-optical properties were modeled as a function of time to account for material degradation and performance changes over the satellite's lifespan. The impact of eclipses on battery performance and thermal behavior was analyzed, ensuring accurate modeling of power and thermal management during shadow periods. The thermal influence of thruster operations was included, capturing the transient heating and cooling effects associated with orbital maneuvers, these phenomena are illustrated on Figure 8. The complete simulation representing the lifespan of 17 years was completed in **5 days**, achieving more than a **x1000 real-time acceleration factor**. This performance highlights the efficiency of the *SIM* and *TCS* simulators in handling large-scale, computationally intensive tasks.

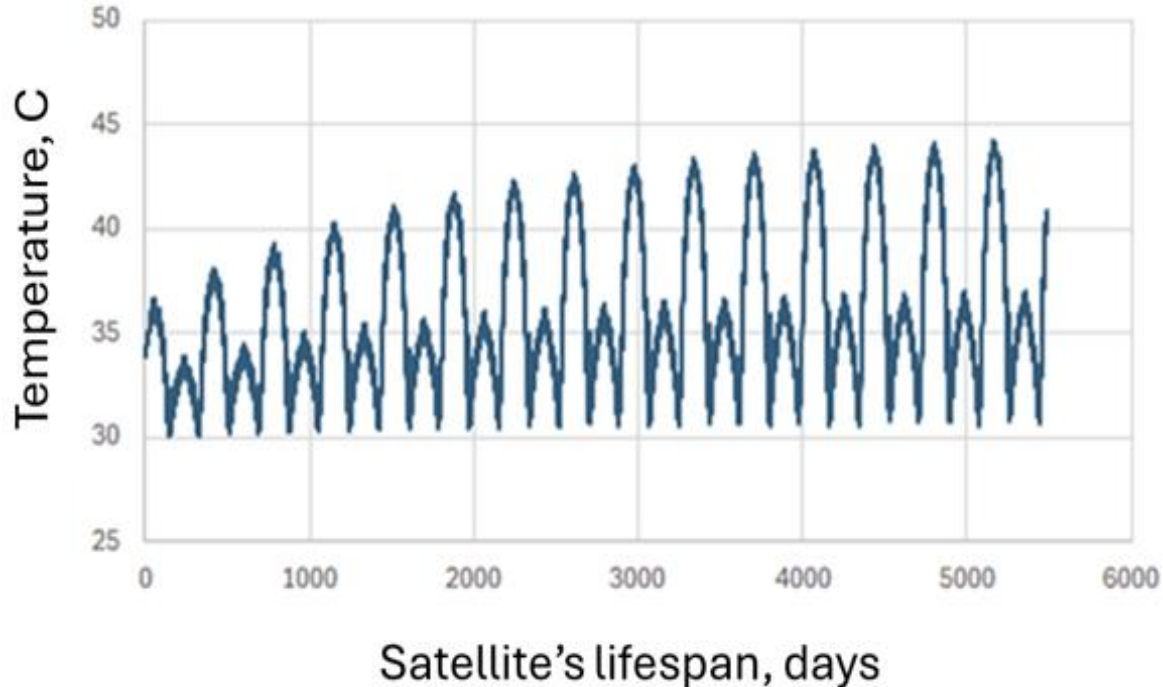


Figure 8. Average Temperature of a Node of the TMM for the Whole Satellite Lifecycle. *The aging effect of materials is depicted.*

The TMM used in this study included MPL and 5,900 nodes. The simulation provided valuable insights into the satellite's thermal performance and reliability over its operational lifespan, for instance the identification of potential thermal risks and anomalies, paving the way to proactive mitigation strategies. Furthermore, the simulation allowed the validation of the satellite's thermal design under realistic, long-duration operational scenarios, as well as the demonstration of the *SIM* and *TCS* simulators' ability to handle multi-year simulations efficiently.

VI. Conclusion and Future Work

A. Conclusion

This study has addressed several critical technical challenges, allowing to reach a higher TRL on this topic, while also identifying areas for further improvement, paving the way for more efficient and accurate thermal management in future space missions.

The study has proven that TMM derived from PSR can be effectively used for real-time thermal simulations. This ensures that simulations are based on highly correlated and accurate models, enhancing their reliability. The ability to compare telemetry data with simulation results in real time has been clearly demonstrated. This capability is crucial for anomaly detection, diagnosis, and operational decision-making. The *SIM* simulator's ability to modify input characteristics “on the fly” during simulation provides unprecedented flexibility, enabling operators to adapt to changing conditions and test various scenarios without interrupting the simulation.

The high modularity and advanced features of *SIM* have made it a viable replacement for the legacy *OBS* system, offering improved performance and adaptability for modern satellite operations.

While the study has achieved significant milestones, some technical challenges remain to be addressed, for instance the original TMM, in its raw form, is not fully optimized for real-time simulations. Further preparation and optimization of both TMM and GMM are required to ensure their suitability for real-time applications. The *Dissipator* module, which calculates equipment dissipations, needs to be extended to account for additional factors such as satellite mode, unit mode, and equipment status. This enhancement will improve the accuracy of dissipation estimates and, consequently, the overall fidelity of the thermal simulations.

B. Future Work

Building on the current proof of concept, initially introduced at ESTEW2024¹⁰, several key areas have been identified to further enhance the usability and effectiveness of rationalized thermal simulators. In operations, improvements will focus on refining the real-time simulation framework to better support mission needs. For instance a better estimation of the dissipations through satellite's modes and submodes given by TM, as well as the validation of modes and submodes through equipments intensities. Also the deduction of the inertia of the satellite, meaning that a substantial number of orbits have to be simulated in order to fit with TM data (“only” 300 TM for 6000 nodes for the TMM), achieving an accurate reproduction of the satellite's dynamics presents a significant challenge. During the design phase, enhancing the preliminary stage will allow for better orbit optimization estimations, including solar array maximization, battery management, unit positioning, radiator sizing, and aging effects based on in-flight experience. In the AIT phase, the exploitation of currently unused data repositories between thermal balance tests will enable better test sequencing and assist in TMM correlation. For the LEOP, a more accurate estimation of the thermal regulation power required during LEOP and EOR will be developed, incorporating solar array and battery capabilities. Additionally, advancements in software modularity will facilitate the integration of cloud computing and AI-driven solutions for fault detection, diagnosis¹¹, and predictive maintenance¹². By addressing these next steps, this work aims to further optimize thermal simulation processes, ensuring adaptability to future missions and seamless integration into a broader digital space operations ecosystem.

In view of everything that has been described above, the road map, associated with this first step, has been divided into 5 phases, see Figure 9. Regarding the first phase, its purpose is to determine the feasibility of this concept by

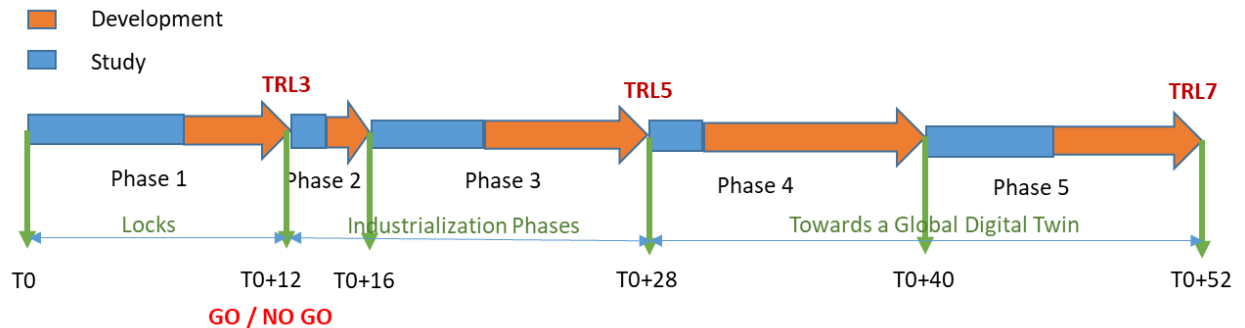


Figure 9. Roadmap Regarding Thermal Digital Twin.

lifting or not the associated technological locks, being the acquisition/estimation of the satellite's dissipations in real

time, the initialization of the thermal model (which cannot be done from the simple reading of the flight TM) and the accessibility to the TM (either on a given date or during a real-time simulation). A second phase of the thermal digital twin should improve its representativeness in terms of material aging and the characterization of albedo and terrestrial fluxes according to the latitude and longitude of the satellite. At the end of this phase, there shall be sufficient progress on the representativeness of the digital twin to consolidate and industrialize the tool. Regarding the third phase, a first part will have to concern the architecture of the tool, its virtualization and its security (cyber rules), this should initially make the validation phases more fluid and subsequently the use phases in Expert and User modes. A second part will be dedicated to the development of a function to verify the correct thermal loading of the P/L according to the power available at the SA and battery level. A fourth phase should make it possible to move towards a global digital twin by integrating this thermal digital twin into the operations simulator in order to have a better representation of all the other subsystems. Finally, the last phase is necessary to provide this digital twin with the autonomy for each subsystem, to diagnose and anticipate future anomalies in order to propose the appropriate corrective sequences. These corrective sequences will come from a state of the art and will be able to notably rely on AI.

References

- ¹Shin, S., Lim, J. H., & Kim, C. G. (2024). Thermal parameter determination in underdetermined spacecraft thermal models using surrogate modeling. *Acta Astronautica*, 214, 583-592.
- ²Xiong, Y., Guo, L., Zhang, Y., Xu, M., Tian, D., & Li, M. (2022). Surrogate modeling for spacecraft thermophysical models using deep learning. *Neural Computing and Applications*, 34(19), 16577-16603.
- ³Tanaka, H., & Nagai, H. (2023). Thermal surrogate model for spacecraft systems using physics-informed machine learning with POD data reduction. *International Journal of Heat and Mass Transfer*, 213, 124336.
- ⁴Hugonnot, P., Basset, T., Connil, P., Brunetti, F., & Ferrier, M. (2016, July). TMRT (Thermal Model Reduction Tool): Presentation of the tool and application on satellite model reduction for launcher coupled analysis. 46th International Conference on Environmental Systems.
- ⁵Bernard, M., Basset, T., & Etchells, J. (2010). Thermal model reduction tool. *24th Eur. Work. Therm. ECLS Softw., ESA/ESTEC, Noordwijk, The Netherlands*, 169-180.
- ⁶Arowoia, V. A., Moehler, R. C., & Fang, Y. (2024). Digital twin technology for thermal comfort and energy efficiency in buildings: A state-of-the-art and future directions. *Energy and Built Environment*, 5(5), 641-656.
- ⁷Lei, Z., Zhou, H., Hu, W., Liu, G. P., Guan, S., & Feng, X. (2021). Toward a web-based digital twin thermal power plant. *IEEE Transactions on Industrial Informatics*, 18(3), 1716-1725.
- ⁸Vadez, V. (2022). *Geometric simplification for radiative thermal simulation of satellites* (Doctoral dissertation, Université Côte d'Azur).
- ⁹Meseguer, J., Pérez-Grande, I., & Sanz-Andrés, A. (2012). *Spacecraft thermal control*. Elsevier.
- ¹⁰Darrau, A., Andre M., Basset, T., & Brunetti, F. (2024). Rationalization of Thermal Simulators for Operations. *38th Eur. Work. Therm. ECLS Softw., ESA/ESTEC, Noordwijk, The Netherlands*.
- ¹¹Ibrahim, S. K., Ahmed, A., Zeidan, M. A. E., & Ziedan, I. E. (2020). Machine learning techniques for satellite fault diagnosis. *Ain Shams Engineering Journal*, 11(1), 45-56.
- ¹²Ochuba, N. A., Usman, F. O., Okafor, E. S., Akinrinola, O., & Amoo, O. O. (2024). Predictive analytics in the maintenance and reliability of satellite telecommunications infrastructure: a conceptual review of strategies and technological advancements. *Engineering Science & Technology Journal*, 5(3), 704-715.